Supplementary Material

Attention-Aware Task Scheduling via Deep Reinforcement Learning under Partial Observability for Human-Centric Productivity Optimization

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S1 Complete simulator specification

S1.1 Episode timeline

Each episode (workday) is generated from a single integer seed that fixes, through independent draws from one pseudo-random generator: the arrival slots, difficulties and slack of all tasks; the initial state (H_0, F_0); the process-noise sequence $\xi_{0:T-1}$; and the observation-noise sequence $\epsilon_{0:T}$. Because the noise sequences are pre-drawn, two policies evaluated on the same seed experience the same exogenous randomness (common random numbers), and their outcomes differ only through their decisions. Within slot t:

1. Tasks with $e_i \leq t$ become available.
2. The scheduler receives the 13-dimensional observation o_t (main text, Eq. (7)), computed from A_t, F_t, ϵ_t and the queue.
3. The action is mapped to a task (EASY/HARD: EDF within the chosen class, with a fallback to the other class; EDF/SPT: the earliest-deadline or shortest-remaining task overall; main text, Eq. (9)) or to rest.

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4. If working on task i: the remaining work is decreased by ρ_t (Eq. (5)); if it reaches zero, the task is completed at $t+1$ and the task reward is credited; H and F are updated with the work branch of Eqs. (3)–(4). If resting: the rest branch is applied.

5. Process noise is added to H, and both states are clipped to $[0, 1]$.

6. The cognitive reward $\mu A_{t+1} - \lambda F_{t+1}$ is added, and every unfinished task whose deadline equals $t+1$ incurs the miss penalty (once).

S1.2 Productivity function

Figure S1 shows the work rate $\rho = \rho_0 A^{0.5+d}(1 - \kappa F)$ for representative difficulties and fatigue levels. At $A = 0.9$ and $F = 0.1$, an easy task ($d = 0.2$) proceeds at 0.88 units/slot and a hard task ($d = 1$) at 0.81. At $A = 0.4$ and $F = 0.8$, the rates fall to 0.32 and 0.15, respectively. Hard work is thus disproportionately penalised in low-attention states, which creates the incentive to match task difficulty to cognitive state.

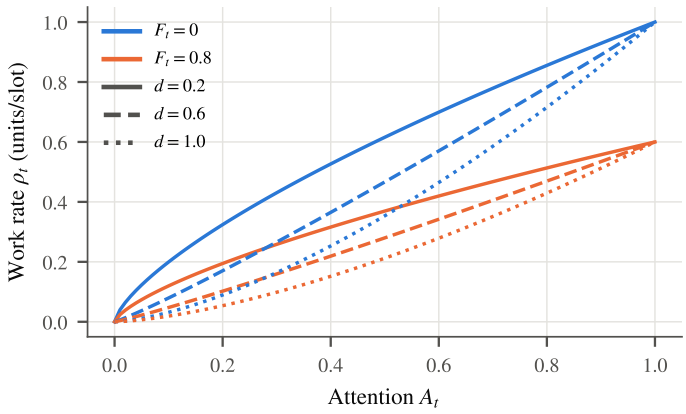


Figure S1: Work rate as a function of attention for three difficulty levels and two fatigue levels.

S1.3 Environment variants used in the ablations

- **No observation noise**: $\sigma_o = 0$; the cognitive state is observed exactly (an MDP with respect to the cognitive variables, although future arrivals remain hidden).
- **No circadian process**: $\gamma = 0$, so that $A_t = H_t$.
- **Static capacity (no cognition)**: $A_t \equiv 0.60$ and $F_t \equiv 0.30$ at all times. The resulting work rate of $0.60^{0.5+d} \times 0.85$ (0.48 units/slot for $d = 0.6$) delivers a daily throughput comparable to the break-taking heuristics in the full model, so the variant removes *dynamics* without drastically changing *capacity*. Breaks have no benefit in this variant.
- **Task-only reward**: $\lambda = \mu = 0$; the dynamics are unchanged.
- **Worker profiles**: resilient ($\alpha, \delta \times 0.7$; $\beta, \eta \times 1.3$) and fatigue-prone ($\alpha, \delta \times 1.3$; $\beta, \eta \times 0.75$).
- **Workload**: $N = 8$ (4 initial) and $N = 16$ (8 initial) tasks per day.

S2 Reproducibility

All experiments can be reproduced with the released code:

```
pip install -r requirements.txt
python -m pytest # simulator unit tests
python -m experiments.run_eval heuristics # tune + test rule-based baselines
```

```
python -m experiments.run_rl --workers 4 # train all 82 agents
python -m experiments.run_eval robustness # zero-shot robustness study
python -m experiments.run_eval traces # behavioural traces
python -m analysis.analyze # all tables, figures, statistics
```

Seed ranges: training days = $s \times 1\,000\,003 + e$ for training seed s and episode e; validation days $10^7, \ldots, 10^7 + 99$; test days $2 \times 10^7, \ldots, 2 \times 10^7 + 499$; tuning days $3 \times 10^7, \ldots, 3 \times 10^7 + 299$. One training run takes about 720 s (AA-DQN) on a single CPU thread.

S3 Baseline tuning

Table S1 lists the parameters selected by grid search for each environment variant (maximum mean return on 300 tuning days). "off" indicates that the fatigue threshold was never triggered ($\tau_F = 1.01$). In the static-capacity variant, rest has no benefit, and all threshold rules learn to (almost) never rest.

Table S1: Tuned heuristic parameters per environment variant.

Variant	EDF-Periodic (p)	EDF-Threshold (τ_F, τ_A)	Cog-Heuristic (τ_F, τ_A, τ_H)
Full model	4	(0.6, 0.7)	(0.4, 0.6, 0.9)
No observation noise	4	(0.3, 0.7)	(0.3, 0.6, 0.9)
No circadian process	4	(0.4, 0.7)	(0.4, 0.6, 0.9)
Static capacity	10	(0.7, 0)	(0.7, 0, 0.8)
Task-only reward	4	(0.7, 0.7)	(0.5, 0.6, 0.9)
$\lambda = 0.05$	4	(0.6, 0.7)	(0.4, 0.6, 0.9)
$\lambda = 0.2$	3	(0.4, 0.7)	(0.3, 0.6, 0.9)
$\lambda = 0.4$	2	(0.3, 0.7)	(0.3, 0.6, 0.9)

S4 Checkpoint selection

Table S2 compares, for each learned policy, the validation-selected checkpoint (primary results) with the final iterate after 5000 episodes. Selection improves both DQN-based agents, which is consistent with the policy oscillation visible in Fig. S2. For AA-PPO, whose learning curve is smooth, the final iterate is practically as good as the selected checkpoint. The ranking AA-PPO > AA-DQN > DQN holds for both checkpoint choices.

Table S2: Test performance (500 days) of validation-selected versus final-iterate checkpoints (mean ± s.d. over ten seeds).

Method	Checkpoint	Return	On-time rate	Mean selected episode
DQN	validation-selected	10.29 ± 0.26	0.677 ± 0.008	3490
DQN	final iterate	9.84 ± 0.34	0.663 ± 0.011	5000
AA-DQN	validation-selected	12.17 ± 0.15	0.728 ± 0.005	3350
AA-DQN	final iterate	11.55 ± 0.35	0.710 ± 0.013	5000
AA-PPO	validation-selected	13.04 ± 0.30	0.751 ± 0.013	4810
AA-PPO	final iterate	13.02 ± 0.27	0.751 ± 0.012	5000
AA-PPO, intensity	validation-selected	13.20 ± 0.10	0.754 ± 0.003	4540
AA-PPO, intensity	final iterate	13.18 ± 0.09	0.755 ± 0.003	5000

S5 Statistical comparisons for AA-DQN

Table S3 gives the paired comparisons of the hybrid AA-DQN with each rule-based baseline, in the same format as the AA-PPO comparisons in the main text (Table 5).

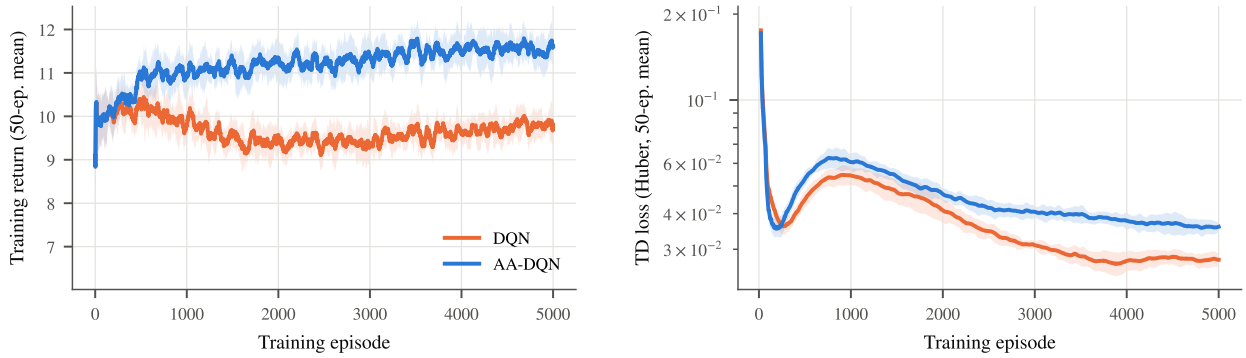


Figure S2: Training return (left; 50-episode moving average, ϵ-greedy) and Huber TD loss (right) for DQN and AA-DQN (mean ± s.d. over ten seeds).

Table S3: Paired comparison of AA-DQN (hybrid actions) with each baseline on 500 common-random-number test days: mean difference with 95% bootstrap CI (on-time rates in percentage points), matched-pairs rank-biserial correlation r_{rb} (positive favours AA-DQN), and Holm-adjusted Wilcoxon signed-rank p-values over the nine comparisons.

	Episode return			On-time rate (pp)			Weighted on-time rate (pp)		
AA-DQN vs.	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}
Random	+3.32 [+3.12, +3.52]	0.97	$< 10^{-4}$	+19.3 [+18.3, +20.3]	1.00	$< 10^{-4}$	+16.7 [+15.7, +17.7]	0.98	$< 10^{-4}$
FIFO	+8.68 [+8.47, +8.91]	1.00	$< 10^{-4}$	+29.6 [+28.7, +30.5]	1.00	$< 10^{-4}$	+23.3 [+22.5, +24.0]	1.00	$< 10^{-4}$
EDF	+7.93 [+7.70, +8.17]	1.00	$< 10^{-4}$	+26.0 [+25.1, +26.9]	1.00	$< 10^{-4}$	+20.0 [+19.3, +20.7]	1.00	$< 10^{-4}$
SPT	+4.22 [+4.06, +4.39]	1.00	$< 10^{-4}$	+10.2 [+9.6, +10.9]	0.96	$< 10^{-4}$	+16.1 [+15.3, +16.8]	1.00	$< 10^{-4}$
EDF-Periodic	+0.76 [+0.58, +0.94]	0.34	$< 10^{-4}$	+6.7 [+5.9, +7.6]	0.69	$< 10^{-4}$	-0.2 [-0.9, +0.5]	-0.05	0.6403
EDF-Threshold	+0.82 [+0.64, +1.01]	0.36	$< 10^{-4}$	+7.4 [+6.5, +8.3]	0.72	$< 10^{-4}$	+0.5 [-0.3, +1.2]	0.01	0.8269
SPT-Periodic	-0.08 [-0.24, +0.08]	-0.06	0.2860	+1.9 [+1.2, +2.6]	0.24	$< 10^{-4}$	+6.3 [+5.5, +7.2]	0.62	$< 10^{-4}$
SPT-Threshold	-0.13 [-0.29, +0.02]	-0.08	0.2390	+1.4 [+0.7, +2.1]	0.20	0.0002	+5.7 [+4.9, +6.6]	0.61	$< 10^{-4}$
Cog-Heuristic	+1.58 [+1.42, +1.72]	0.80	$< 10^{-4}$	+8.9 [+8.3, +9.6]	0.92	$< 10^{-4}$	+12.2 [+11.4, +13.1]	0.93	$< 10^{-4}$

S6 Component and sensitivity analysis

Table S4 reproduces the component study of the main text with all configurations, including the history length k and the recurrent (GRU) encoder.

S7 Audit trail of the baseline set and action space

For transparency, we document a design change made during the study. In the first version of the experiments, the agent's action set contained only the intensity actions {EASY, HARD, BREAK} of the original formulation, with EDF dispatch within the chosen class. The rule-based baselines were EDF-based (EDF-Periodic, EDF-Threshold, Cog-Heuristic). A pre-submission review argued that SPT-based rest heuristics should also be included, because SPT was the best classical rule in the static-capacity ablation. When added, SPT-Threshold obtained a higher test return than the intensity-only agent. The reason was structural: that agent could not express SPT sequencing. We therefore adopted the hybrid action set, in which every baseline is representable, added the observation of each class's shortest remaining work, set the training budget to 5000 episodes (the upper end of the range in our original protocol), and re-ran all experiments. To avoid over-fitting the design to the test set, this decision was made once, and no further design changes were made after the final runs were launched. The intensity-only agent trained under the final protocol is reported as a component ablation (Table S4).

Table S4: Component ablation and sensitivity (mean ± s.d. over seeds; 500 test days).

Configuration	Seeds	Return	On-time rate	Train time (s)
AA-DQN (hybrid actions, k=4, Double, Dueling)	10	12.17 ± 0.15	0.728 ± 0.005	720
– without history (k=1)	3	10.34 ± 0.19	0.683 ± 0.010	551
– without Double Q-learning	3	12.07 ± 0.19	0.720 ± 0.014	552
– without dueling head	3	12.28 ± 0.04	0.725 ± 0.004	517
– intensity-only actions {easy, hard, break}	10	12.61 ± 0.18	0.740 ± 0.005	582
history k=2	3	11.17 ± 0.12	0.701 ± 0.001	568
history k=8	3	12.48 ± 0.37	0.733 ± 0.011	605
GRU encoder over k=8 history	3	12.54 ± 0.16	0.728 ± 0.006	2463
learning rate 10^{-4} (nominal 5×10^{-4})	3	11.75 ± 0.16	0.719 ± 0.007	646
learning rate 10^{-3}	3	12.19 ± 0.17	0.726 ± 0.001	723
discount $\gamma_{\mathrm{RL}} = 0.95$ (nominal 0.99)	3	11.72 ± 0.29	0.712 ± 0.009	664
discount $\gamma_{\mathrm{RL}} = 0.999$	3	12.24 ± 0.08	0.727 ± 0.001	659
Vanilla DQN (k=1, no Double/Dueling)	10	10.29 ± 0.26	0.677 ± 0.008	614
AA-PPO (hybrid actions)	10	13.04 ± 0.30	0.751 ± 0.013	171
AA-PPO (intensity actions)	10	13.20 ± 0.10	0.754 ± 0.003	172

S8 Full robustness results

Table S5 reports the return and on-time rate of every method under each zero-shot test condition. Learned policies are averaged over training seeds, and heuristics use the parameters tuned in the nominal environment.

Table S5: Zero-shot robustness: return / on-time rate on 500 test days per condition. Bold marks the best return.

Test condition	AA-PPO	AA-PPO, intensity	AA-DQN	AA-DQN, intensity	AA-DQN, randomised	DQN	SPT-Threshold	SPT-Periodic	EDF-Periodic
$\sigma_o = 0$	13.04 / 0.754	**13.24 / 0.758**	12.14 / 0.728	12.57 / 0.740	11.59 / 0.714	10.00 / 0.669	11.96 / 0.714	12.25 / 0.709	11.41 / 0.660
Nominal ($\sigma_o = 0.1$)	13.04 / 0.751	**13.20 / 0.754**	12.17 / 0.728	12.61 / 0.740	11.63 / 0.714	10.29 / 0.677	12.30 / 0.714	12.25 / 0.709	11.41 / 0.660
$\sigma_o = 0.2$	13.03 / 0.744	**13.04 / 0.742**	12.20 / 0.726	12.68 / 0.739	11.67 / 0.713	10.68 / 0.686	12.04 / 0.692	12.25 / 0.709	11.41 / 0.660
$\sigma_o = 0.3$	**12.76 / 0.725**	12.72 / 0.723	12.27 / 0.725	12.71 / 0.734	11.72 / 0.713	10.98 / 0.693	11.35 / 0.656	12.25 / 0.709	11.41 / 0.660
Resilient worker	15.09 / 0.819	**15.46 / 0.831**	14.31 / 0.798	14.76 / 0.811	13.82 / 0.784	12.28 / 0.738	14.12 / 0.767	14.03 / 0.751	15.01 / 0.782
Fatigue-prone worker	**11.04 / 0.682**	10.87 / 0.672	10.05 / 0.656	10.40 / 0.664	9.52 / 0.643	8.48 / 0.619	10.59 / 0.662	10.21 / 0.659	7.46 / 0.529
8 tasks/day	12.83 / 0.964	13.04 / 0.974	11.71 / 0.931	12.20 / 0.947	11.79 / 0.930	9.55 / 0.843	11.99 / 0.899	11.94 / 0.898	**13.06 / 0.973**
16 tasks/day	11.83 / 0.595	**11.83 / 0.590**	9.69 / 0.531	10.03 / 0.533	10.35 / 0.569	9.71 / 0.548	11.61 / 0.587	11.48 / 0.579	7.32 / 0.417

S9 Reward-weight sweep

Table S6: AA-DQN as a function of the fatigue weight λ (attention weight $\mu = \lambda/2$; mean ± s.d. over three seeds).

λ ($\mu = \lambda/2$)	On-time rate	Mean fatigue	High-fatigue exposure	Break share
0	0.708 ± 0.009	0.365 ± 0.015	0.000	0.115
0.05	0.718 ± 0.009	0.354 ± 0.009	0.000	0.124
0.1	0.727 ± 0.005	0.344 ± 0.002	0.000	0.137
0.2	0.731 ± 0.007	0.304 ± 0.010	0.000	0.181
0.4	0.723 ± 0.012	0.225 ± 0.008	0.000	0.304